

15 A
$$\frac{1}{5.0 \times 10^5} \sin \theta_{\text{red}} = (1)(700 \times 10^{-9}) \Rightarrow \theta_{\text{red}} = 20.49^\circ$$

$$\frac{1}{5.0 \times 10^5} \sin \theta_{\text{blue}} = (1)(400 \times 10^{-9}) \Rightarrow \theta_{\text{blue}} = 11.54^\circ$$

$$\Delta \theta = 8.95 = 9.0^\circ$$

16 A Constant pressure compression – in the p - T diagram it is still a horizontal line $pv = nRT$